
PhysWM: Grounding Predictive World Models in Physical Equations

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Abstract

1 Predictive world models can forecast dynamics without organizing their represen-
2 tations around the physical variables governing a system. We introduce PHYSWM,
3 a two-path architecture that grounds an action-conditioned predictive world model
4 through known physical equations. A neural path predicts the next state from data,
5 while an equation path maps the same predictive latent through a low-capacity
6 probe to physical variables and a frozen differentiable solver. Without parameter
7 labels, the equation path learns to reproduce the neural path’s stopped-gradient
8 prediction. We evaluate the method on FetchPush, randomized PushT, and Cart-
9 Pole. On FetchPush, the neural path beats persistence in every seed and equation
10 grounding reduces teacher-matching error from 8.758 to 0.752. On randomized
11 PushT, inferred effective dynamics beat shuffled episode assignments in all three
12 seeds. FetchPush exposes a solver-adequacy failure, while the small CartPole runs
13 remain below the data floor. These results support equation-compatible effective
14 coordinates without equating them with uniquely recovered material parameters.

15 1 Introduction

16 World models compress observations and actions into representations useful for predicting future
17 dynamics (4; 5; 6; 11). Predictive accuracy alone, however, does not ensure that these representations
18 expose mass, damping, stiffness, friction, or control response. A high-capacity predictor may exploit
19 trajectory correlations while leaving the governing physical factors entangled, even when its next-state
20 predictions are accurate (9). This separates predicting *what happens* from representing *why the system*
21 *evolves that way*.

22 We study whether known physical equations can ground a flexible predictive world model without
23 physical-parameter annotations. PHYSWM produces two next-state predictions from the same action-
24 conditioned latent (Figure 2). Path A is a learned decoder trained on observed transitions. Path B
25 maps the latent to a compact vector $\hat{\theta}$ and evaluates a frozen differentiable solver. Path B targets
26 Path A’s stopped-gradient prediction rather than the dataset state directly. The neural path remains
27 anchored to data and free to absorb model mismatch; the physical path asks whether the model’s
28 predictive belief admits a compact equation-based explanation. This fixed-decoder role is related
29 to differentiable system identification (8), but our training target is the predictor’s belief rather than
30 rendered pixels or labeled physical parameters.

31 Our contribution is threefold. First, we introduce a shared-latent, two-path objective for grounding
32 predictions in a supplied equation family without parameter labels. Second, we evaluate grounding
33 functionally by changing only the solver’s parameter vector while holding states, actions, weights,
34 and equations fixed. Third, we distinguish *semantic parameters*, which match named ground-truth
35 quantities, from *effective physical coordinates*, which improve equation rollouts but may compensate
36 for non-identifiability or solver mismatch.

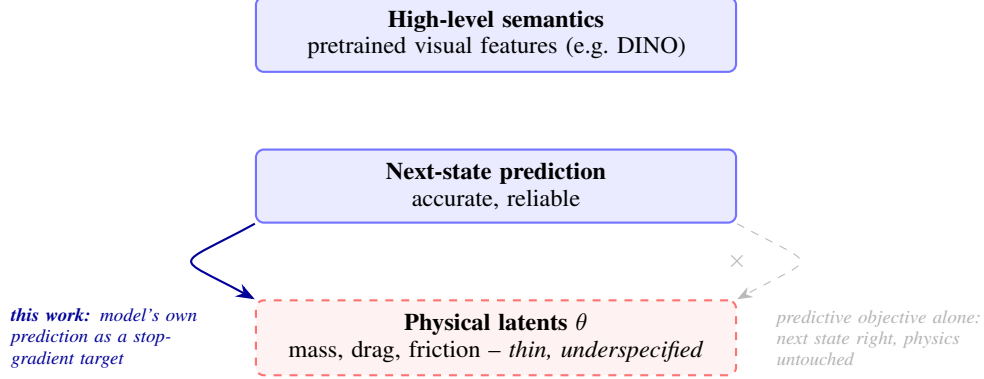


Figure 1: **Filling a thin foundation from the top.** Self-supervised encoders build a hierarchy that is tall at the top (rich semantics) but thin at the bottom (physical structure). A world model trained only for next-state prediction inherits this shape: it gets the top level right without the bottom level ever becoming physically meaningful. We use the reliable top-level signal, the model’s own next-state prediction, as the training target for a low-capacity, physics-constrained probe at the bottom, without ever touching a physical-parameter label.

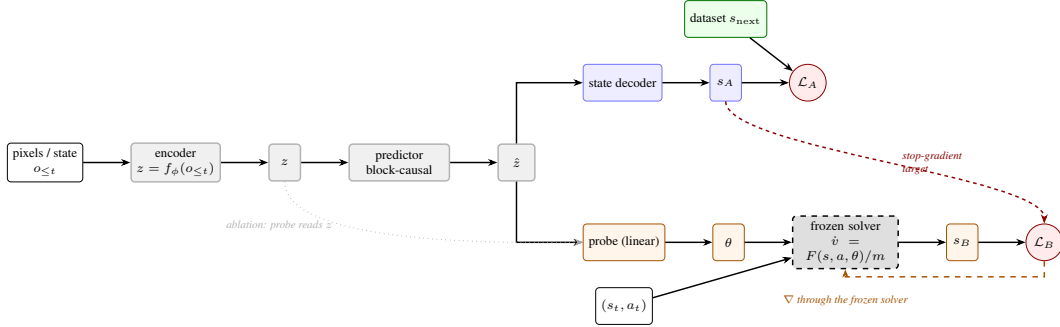


Figure 2: **PHYSWM architecture.** Encoder and predictor form one shared trunk producing a single action-conditioned latent $\hat{z}$; both heads read $\hat{z}$, never a private view of the observation. Path A (blue) is the ordinary predictor: a decoder reads $\hat{z}$ to a state s_A , and $\mathcal{L}_A$ (Equation (4)) regresses the dataset’s ground truth, exactly what a predictive world model already does. Path B (orange) reads a physics vector θ off the same $\hat{z}$ with a deliberately low-capacity probe, integrates it through a *frozen, zero-parameter* solver (gray, dashed) to get an independent estimate s_B . $\mathcal{L}_B$ does *not* use the dataset label: it regresses s_A with gradient stopped (red, dashed), so the only route back to a learnable parameter is through the fixed solver into θ (orange dashed arrow), and θ can only improve by becoming a better physical explanation of what Path A already predicts. The dotted arrow is not part of the method: it marks the pre-action routing ablation (Section 3), where the probe reads z instead of $\hat{z}$.

2 Equation-grounded predictive representations

Given observations o_t , actions a_t , and prediction state s_t , an encoder and causal action-conditioned predictor form

$$z_t = E_\phi(o_t), \quad \hat{z}_t = P_\psi(z_{\leq t}, a_{\leq t}). \quad (1)$$

The neural decoder produces $\hat{s}_{t+1}^A = D_\omega(\hat{z}_t)$ and is trained in normalized state space:

$$\mathcal{L}_A = \|N(\hat{s}_{t+1}^A) - N(s_{t+1})\|_2^2. \quad (2)$$

A bounded low-capacity probe pools only causal context latents,

$$\hat{\theta} = \text{Bound}(\rho_\epsilon(\text{Pool}(\{\hat{z}_t\}_{t \in \mathcal{C}}))), \quad (3)$$

and a frozen solver S , containing no learnable parameters, produces $\hat{s}_{t+1}^B = S(s_t, a_t, \hat{\theta})$. The grounding loss is

$$\mathcal{L}_B = \|N(\hat{s}_{t+1}^B) - \text{sg}(N(\hat{s}_{t+1}^A))\|_2^2, \quad \mathcal{L} = \mathcal{L}_A + \alpha \mathcal{L}_B. \quad (4)$$

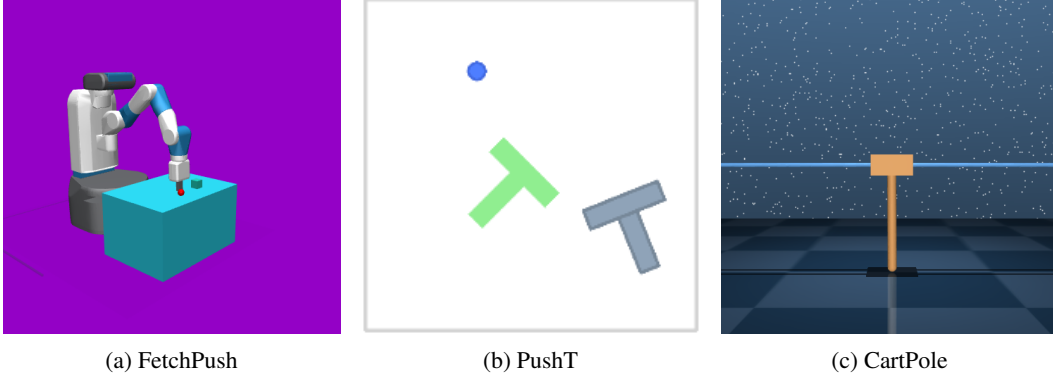


Figure 3: Rendered observations from the three evaluated environments. FetchPush and CartPole use MuJoCo; PushT uses Pymunk contact dynamics.

44 The stopped target prevents $\mathcal{L}_B$ from moving the Path-A decoder through its target edge, while
 45 gradients through S , ρ_ξ , and P_ψ shape the shared predictor. The default probe is linear, preventing
 46 it from hiding another world model. $\hat{\theta}$ is always a forward-pass output, never a free test-time
 47 optimization variable. The stop-gradient follows self-distillation methods (7; 1; 3); here it enforces a
 48 one-way dependency between a predictive teacher path and an equation-constrained student path.

49 **Causal evaluation split.** The identifier observes only completed context transitions. Metrics are
 50 computed on later query transitions, and no query outcome is accessible to the probe. With the default
 51 eight-frame window, four completed transitions form the context. Functional evaluation uses eight
 52 context transitions followed by seven held-out rollout transitions.

53 **Grounding criterion.** We call a coordinate physically grounded when reading it from the predictive
 54 latent allows the fixed equations to explain held-out dynamics better than a coordinate carrying no
 55 episode-specific information. This is deliberately weaker than claiming recovery of a unique material
 56 parameter: trajectories may identify only ratios such as stiffness/mass, and approximate solvers may
 57 induce compensating effective values.

58 3 Evaluation

59 We test four complementary properties. (1) *Predictive adequacy*: Path A must beat persistence,
 60 $\hat{s}_{t+1} = s_t$. (2) *Semantic readout*: where labels exist, a supervised linear probe measures held-out
 61 episode-level R^2 . (3) *Parameter substitution*: the solver receives the inferred vector, another episode’s
 62 shuffled vector, nominal constants, or true parameters when available. (4) *Multi-horizon rollout*: the
 63 same substitutions are rolled forward under fixed held-out actions. All errors are scaled per state
 64 dimension.

65 **FetchPush.** We use the Gymnasium-Robotics FetchPush task with per-episode object mass and
 66 friction variation. The MuJoCo environment supplies real contact dynamics, while the frozen solver
 67 uses a point-mass, linear-friction approximation. The reported primary runs use 256 episodes and
 68 three seeds.

69 **Randomized PushT.** This planar manipulation benchmark extends the contact-rich Push-T task (2)
 70 with episode-level variation in effective dynamics. Its solver is approximate and ground-truth labels
 71 are unavailable for most coordinates, making inferred-versus-shuffled substitution the primary test.
 72 The reported runs use 512 episodes and three seeds.

73 **CartPole.** The DeepMind Control Suite task (10) uses a six-parameter analytic solver for cart
 74 mass, pole mass, pole half-length, gravity, force gain, and damping. Only small, independently
 75 sampled runs of 8, 32, and 64 episodes are complete, so we treat CartPole as a pipeline and data-floor
 76 diagnostic rather than headline evidence.

Table 1: Mean $\pm$ sample standard deviation over three seeds. RMSE is lower better; R^2 is higher better.

Benchmark	Metric	Baseline / control	PHYSWM
FetchPush	Prediction RMSE	Persistence 0.648 ± 0.037	0.420 ± 0.015
FetchPush	Teacher-match RMSE	Ungrounded 8.758 ± 0.398	0.752 ± 0.036
PushT	Substitution RMSE	Shuffled 0.267 ± 0.013	Inferred 0.208 ± 0.011
CartPole	Substitution RMSE	Shuffled 0.339 ± 0.107	Inferred 0.311 ± 0.155
CartPole	Prediction RMSE	Persistence 0.199	0.792

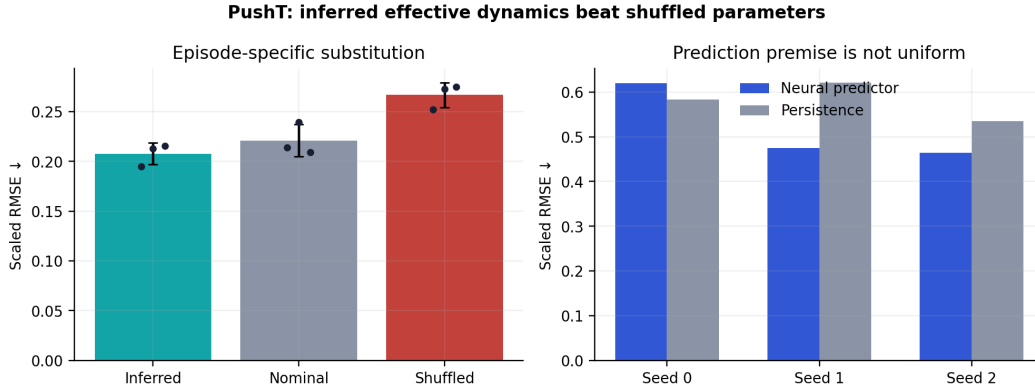


Figure 4: Randomized PushT substitution test over three seeds. Inferred episode-specific coordinates beat shuffled assignments in every seed and nominal parameters in two of three. Error bars show sample standard deviation.

4 Results and discussion

FetchPush satisfies the predictive premise. Path A reaches 0.420 ± 0.015 scaled RMSE, compared with 0.648 ± 0.037 for persistence, and wins in every seed (Table 1). The physics loss also changes the equation path substantially: its error against the stopped-gradient teacher falls from 8.758 ± 0.398 without grounding to 0.752 ± 0.036 with grounding. The shared latent therefore supports an equation-constrained reconstruction of the neural prediction, even though the raw mass and friction coordinates are not semantically identifiable at this data budget.

PushT coordinates carry episode-specific dynamics. Inferred variables obtain 0.208 ± 0.011 substitution RMSE versus 0.267 ± 0.013 for shuffled assignments (Fig. 4). They beat shuffled values in all three seeds and nominal values in two of three. Solver sensitivity is concentrated in the agent’s proportional and derivative gains, so this supports effective controller dynamics rather than complete contact-parameter identification. Path A fails to beat persistence in one seed, which limits the strength of this result.

FetchPush coordinates compensate for solver mismatch. The corrected frozen solver fails its adequacy check even after directly fitting a free parameter vector to each episode: fitted solver RMSE is 0.483, compared with 0.411 for persistence. In the functional run, inferred variables produce lower error than true mass and friction (1.535 versus 6.984), while solver-space task success is 0.03125 for every parameter source. The probe is compensating for the solver’s fixed contact and point-mass approximations. We therefore call $\hat{\theta}$ an effective coordinate rather than a recovered material parameter.

CartPole is still data-limited. Across independent 8-, 32-, and 64-episode runs, inferred coordinates beat shuffled coordinates in a majority of seeds at every scale, but neither Path A nor the inferred solver beats its naive baseline. At 64 episodes, probe and shuffled substitution errors are 0.311 ± 0.155 and 0.339 ± 0.107 , while nominal parameters reach 0.007. The non-monotonic Path-A error across budgets rules out a scaling claim from these small runs.

5 Conclusion and limitations

PHYSWM grounds a predictive world model by requiring its shared action-conditioned representation to support a compact explanation through known physical equations. FetchPush establishes the predictive premise and shows that grounding makes the equation path track its teacher. Randomized PushT shows episode-specific information under parameter substitution. Neither result establishes recovery of uniquely named material parameters.

The study is limited to tiny-CNN diagnostics. The FetchPush solver is inadequate even after oracle fitting, PushT uses an approximate disc model for a T-shaped block, and CartPole has only small independent samples. The method does not discover equations, select informative actions, or demonstrate closed-loop planning or sim-to-real transfer. Larger CartPole runs, stronger visual encoders, and solvers that pass adequacy checks are required before broader claims.

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